

Predicting Cancer Diagnosis Using Support Vector Machines

National Health Interview Survey(NHIS) 2022 | EDWIN OKWOR | SEATTLE UNIVERSITY



According to the World Health Organization, Cancer is the leading cause of death worldwide, accounting for about 10 million deaths in 2020. Cancer is a disease in which some of the body's cells grow uncontrollably and spread to other parts of the body. As per Stanford Medicine health care, there are no single cause for cancer. It's often an interaction of many factors together that cause cancer. Some of the risk factors that could contribute include: Lifestyle, family history/inheritance, exposure to certain viruses, and environmental exposures. The main objective of this project is to predict whether a person will have cancer or not based on some demographic, body metric, life habit features.

01 Introduction

Can everyday health behaviors predict cancer? Using NHIS 2022 data, this project applies SVM models with three kernel types — linear, radial, and polynomial — to classify whether a respondent has ever been diagnosed with cancer based on demographics and lifestyle habits.

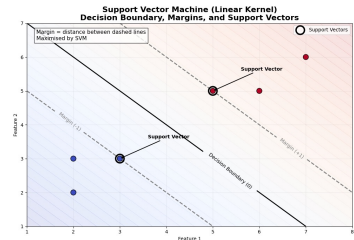
03 TECHNICAL BACKGROUND: Support Vector Machine

Support Vector Machine(SVM) is a supervised machine learning algorithm used for classification and regression tasks. It finds an optimal decision boundary(hyperplane) that separate classes while maximizing the margin between them.

How SVM Works

For linearly separable data, SVM identifies a hyperplane that maximizes the distance(margin) between classes. The margin is defined by key data points called support vectors. For non-linear data, SVM uses kernel functions to map the data into a higher-dimensional space where separation is possible. We optimize(minimize):
$$\frac{1}{2} ||w||^2 + C \sum \xi_i$$

Constraint:
$$y(w \cdot x_i + b) \geq 1 - \xi_i, \xi_i \geq 0$$



Kernels

- Common kernel functions include:
 - Linear kernel: which is suitable for linearly separable data.
$$K(x_i, x_j) = x_i \cdot x_j$$
 - Polynomial kernel: models feature interactions
 - Radial Basis Function(RBF): captures non-linear structure.

Parameters

Parameter	Description	Kernels
C (cost)	Penalty for misclassification; higher C → tighter margin	All
γ (gamma)	Reach of a single training point; high γ → more complex boundary	RBF, Polynomial
d (degree)	Polynomial degree; controls boundary curvature	Polynomial

02 Dataset & Variables

Target: Cancer_Status (ever diagnosed)

Predictors (7):

- Age, Sex, Poverty Ratio
- Weight
- Alcohol days, Soda frequency, Sleep hours

Model Evaluation

The performance any of the SVM models can be assessed using:

- Accuracy:** tells us Out of all predictions, how many are correct? Misleading when working with imbalanced data.
- Precision:** focuses on the event we care about identifying (i.e. Cancer in this project). A high precision indicates that predicted cancer cases are usually correct. Low precision means the model says positive too often even when it is wrong.
- Recall:** focuses on out of the truly positive classes, how many did the model correctly catch? A low recall implies that many positives were missed while a high recall indicates very few false negatives.
- F1-Score:** is a balance between precision and Recall.

Considerations & Limitations

Linear Kernel

- Only works well with classes that are linearly separable
 - Single tuning parameter C — easiest to tune
 - Interpretable
 - Sensitive to class imbalance without weighing
- #### Radial Kernel
- Most flexible — handles complex, non-linear boundaries
 - Two tuning parameters (C, γ) — hence most expensive to tune.
 - High γ — lead to overfitting; Low γ: leads to underfitting
 - Difficult to interpret; misleadingly high accuracy under class imbalance

Polynomial Kernel

- Able to capture feature interactions using degree d
 - Two tuning parameters (C, d) — could lead to overfitting.
 - Less stable than RBF for similar flexibility
 - Two tuning parameters (C, γ) — hence most expensive to tune.
 - High γ — lead to overfitting; Low γ: leads to underfitting
- #### All Kernels
- Feature scaling is required.
 - Class weights or resampling needed for imbalanced data
 - Computationally expensive for large datasets

04 Methodology

Data Source & Target Variable

The dataset was obtained from 2022 National Health Interview Survey(NHIS). The target variable is CANCEREV, converted into a binary outcome: 1 = Cancer, 0 = No Cancer. Invalid or unknown codes (0,7,8,9) were set to NA and those rows dropped (7,501 observations removed).

Feature Selection

- Seven predictors were selected to capture key risk domains:
 - Demographics: Age(years), Sex(male/female),and poverty ratio.
 - Body Metrics: Weight(lbs)
 - Lifestyle habits: Alcohol Consumption(days/year), Soda intake Frequency, and Sleep duration(hours/night).
 - BMI was excluded due to collinearity with Weight.
 - Smoking was not included(34k+ missing due to survey skip-pattern)

Data Preprocessing

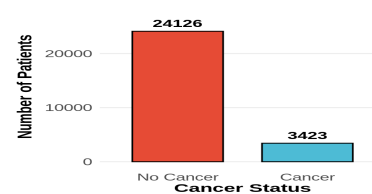
Survey variables from the NHIS dataset were cleaned by recoding standard missing and invalid codes to NA prior to analysis. These codes correspond to survey specific non-response categories such as "refused", "not ascertained", and "unknown". The following recording rules were applied:

- Cancer Status(CANCEREV): 1- Cancer; 2 – No Cancer
- Sex(SEX): values 7-9 set to NA
- Sleep hours(HRSLEEP) values 97 – 99 set to NA.
- For poverty ratio, Age, Weight, Alcohol days(ALCANYNO), Soda Frequency(SODAPNO).
- The NA values were removed.

To Handle Class Imbalance

- The dataset was highly class imbalanced(~ 88% No Cancer vs. ~12% Cancer)
- Class Weights Applied: Cancer = 7, No Cancer = 1 (weight = max count/minority count)

Distribution of Cancer Status



- Age & Sex(<70 missing): rows dropped
- Weight, Alcohol, Soda, Sleep (moderate missingness): median imputation applied – robust to right skewed distribution

Hyperparameter Tuning

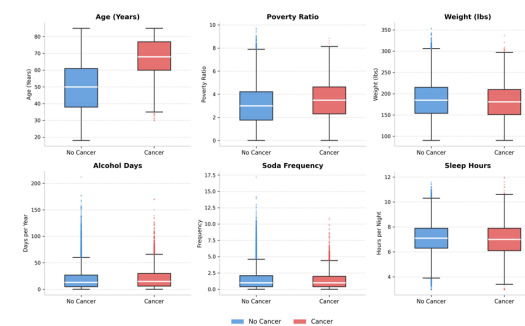
- Cross-validation on training set only.
- Cost ∈ {0.01, 0.1, 1, 10} — all kernels
- Gamma ∈ {0.01, 0.1, 1, 10} — radial kernel
- Degree ∈ {2, 3, 4} — polynomial kernel
- Best params → models refit on full training set before test evaluation

Evaluation Metrics

- Accuracy, Precision, Recall, F1-Score
- Recall = primary metric (medical screening context; missing a cancer case is the costliest error)
- Accuracy noted as misleading under class imbalance

05 Results

Predictor Distributions by Cancer Status | NHIS 2022



07 Conclusion

- Age emerged as the sole dominant predictor of cancer diagnosis across all SVM kernels, with the decision boundary consistently falling between ages 55–65. Lifestyle variables — alcohol, soda, sleep, and weight — showed minimal individual discriminatory power, suggesting these factors cannot predict cancer status in isolation.
- The polynomial kernel (F1 = 0.36, Recall = 0.74) provided the best balance for this imbalanced classification task. The linear kernel (Recall = 0.80) remains preferable in a strict cancer screening context where missing a true case carries the highest cost. The radial kernel, despite its high accuracy, is not recommended without additional tuning to address class imbalance.
- For policymakers, age-targeted screening programs — particularly for individuals 60+ — represent the most evidence-supported intervention from this analysis.

- <https://www.kaggle.com/edwinokwor/nhls-support-vector-machine-algorithm>
- <https://www.kaggle.com/edwinokwor/nhls-support-vector-machine-algorithm>
- <https://www.kaggle.com/edwinokwor/nhls-support-vector-machine-algorithm>

[1] Lynn A. Blewett, Julia A. Rivera Drew, Miriam L. King, Kari C.W. Williams, Daniel Bachman, Annie Chen, and Stephanie Richards. IPUMS Health Surveys: National Health Interview Survey, Version 7.4 [dataset]. Minneapolis, MN: IPUMS, 2024. <https://doi.org/10.18128/0000/7.4> [Links to an external site](https://www.nhls.gov/nhls/links-to-an-external-site) <https://www.nhls.gov/nhls/links-to-an-external-site>

<https://medium.com/@edwinokwor/advanced-data-science-support-vector-machines-an-intuitive-explanation-108a6c720106>